

ECO 5325: Lecture 1

Introduction to International Economics

Philip Economides

Spring 2026

Studying the Global Economy

Course Objectives

For this class we will

- study workhorse models explaining why countries/firms trade
- analyze theoretical and empirical applications of these models
- think critically about the current state of the literature
- develop original research ideas

Synergies with other fields...



Today's Plan

1. Course Logistics
2. About Me
3. Key facts about international trade
4. If time allows.. overview of key models of trade

Course logistics

Housekeeping:

- Office hours: before class at 1pm & by appointment
- No required textbook → journal articles
- Required readings + class participation
- No exams → focus on critical thinking + presentation/writing skills

Content will be available on Canvas.

Course logistics

Content and grading:

- **W1-W6:** Theory of Trade
- **W6-W11:** Trade Policy and Multinationals
- **W11-W16:** Global Finance
- 15% - Class Participation & Paper Critiques ×2
- 20% - Written Assignments ×2
- 10% - Term Paper Proposal
- 35% - Term Paper
- 20% - Class Presentations ×2

Will provide guidance on preparing written assignments. Questions always welcome.

Today's Plan

1. Course Logistics
2. **About Me**
3. Key facts about international trade
4. If time allows.. overview of key models of trade

About Me

Philip Economides

- 2017-2019: RA at ESRI/Trinity College Dublin (Ireland),
2019-2024: PhD Candidate & Lecturer (University of Oregon),
2024-Present: Assistant Professor (Texas Tech University)
- Focus: International Trade, Transport Economics, Applied Econometrics

Where can you find me?

- Holden Hall 236, Tues/Thurs 13:00-14:00, or by appointment
- Email: peconomi@ttu.edu (*use ECO 5325 in the subject line*)

About Me

- Empirical/Structural Work
- Main Research Areas:
 - Role of transportation services in international trade
 - Multinational profit-shifting & activities
 - Environmental implications of international trade

Research Topic I: Trade/Transport

Trade relies heavily on containerships (60-70% of global trade volume)

- Asymmetric demand across bilateral trade routes
- Net outflows of loaded containers (e.g. US-China)
- Empty container repositioning $\implies$ **endogenous transport costs**
- How does government intervention (e.g. OSRA 2022) influence trade outcomes?

Challenges: Data availability (\$), complex theoretical models

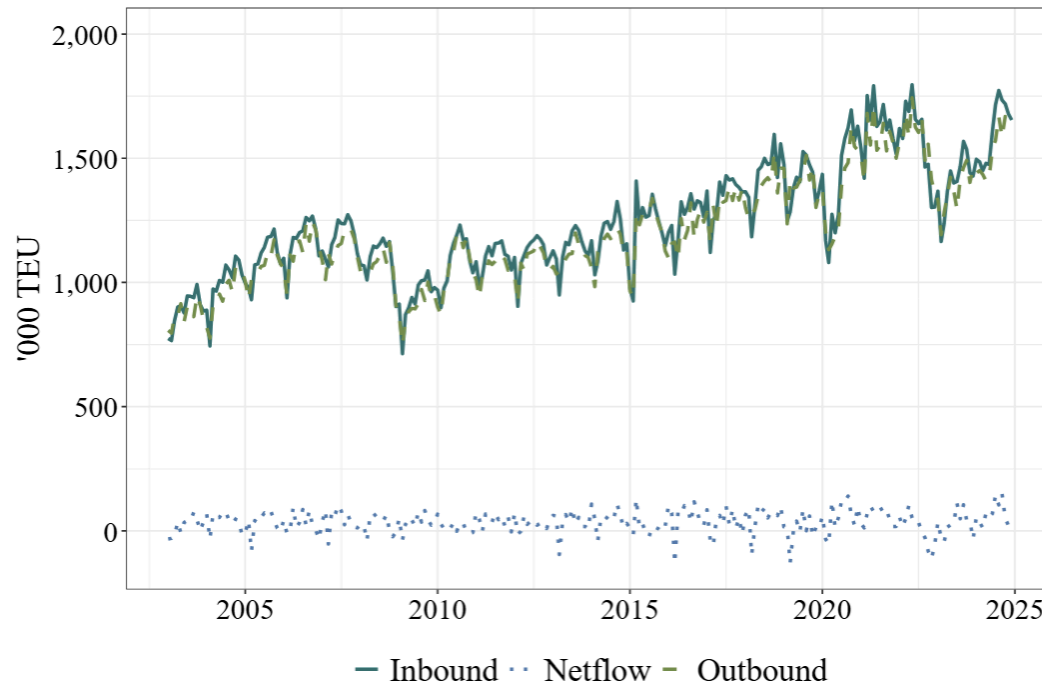


Figure 1. Inbound and Outbound Total Container Flows

Container exchanges are only balanced when taking empties into account

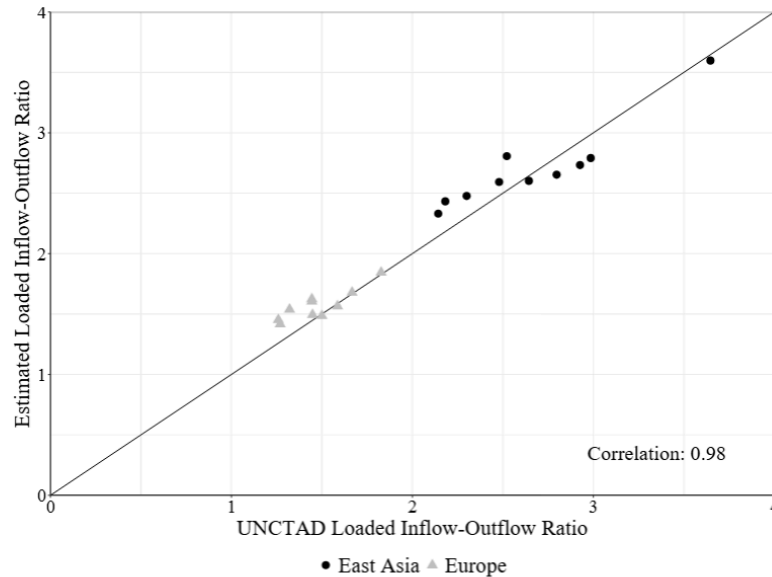


Figure 6. Model Fit – Loaded Container Ratios by Region (2012-2021)

Note: Observed levels originate from UNCTAD records on regional total loaded container flows by year and were untargeted in the estimation of individual country container flow estimates.

Estimates of country-specific container flows fit strongly at observed regional levels

CF: **Moderate restriction to US empty container outflows**

Table. 5. Disaggregated Counterfactual Outcomes

U.S. Measures	Imports	Exports	Imp. Price	Exp. Price	Value	Vol.	Capacity
$\Delta\%$	-17.1	30.0	2.3	-5.7	-8.2	-4.3	-18.5

Note: These results reflect percentage changes from their respective 2017 baseline scenarios of the partial equilibrium model and are based on estimates of loaded container flows & observed levels of associated trade in containerized manufactured goods.

- Cost of servicing assymetric lanes has risen, inducing price response
 - Import leg $\rightarrow$ freight rate increase
 - Export leg $\rightarrow$ freight rate decline
- Imports $\downarrow$, exports $\uparrow \implies$ overall trade falls
- Pronounced decline in trade with East Asian trade partners

Research Topics II: Multinational Firms

Developments in transportation and communication...

⇒ fragmentation of production across national borders

⇒ growth in multinational corporations

Conventional main questions:

- Why do some firms own subsidiaries abroad?
- Which factors determine the location of foreign subsidiaries?
- How is production and employment best organized across locations?

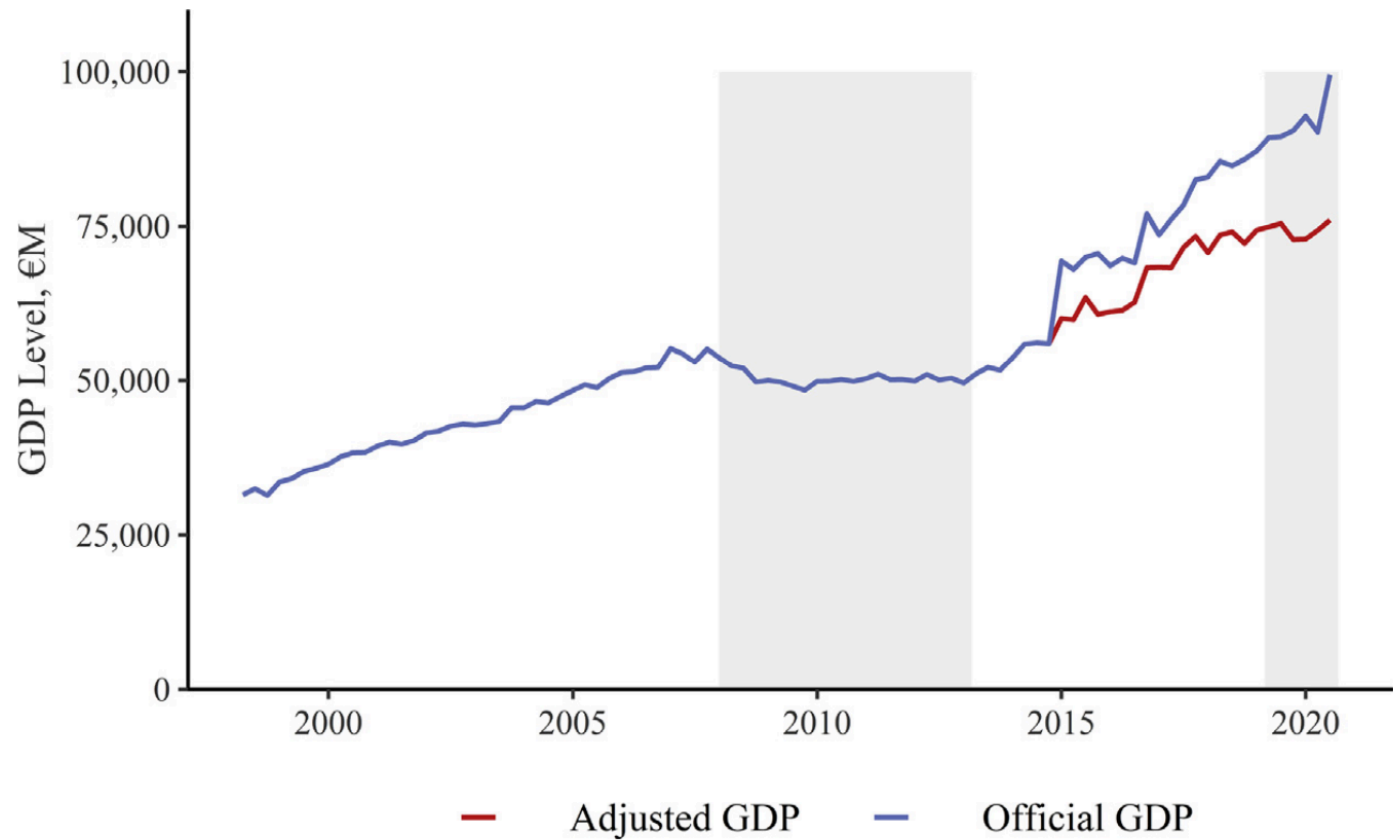
Research Topics II: Multinational Firms

Growing focus on MNEs placing assets in low-tax jurisdictions

“Measuring Economic Activity in the Presence of Superstar MNEs”

(w. G. Nikolaishvili) **Economics Letters, 2023. Volume 226, 111077**

- How do these large shifts in strategic locations of ownership of intellectual property influence domestic economies?
- Main finding – MNE presence in Ireland has inflated GDP by 22% relative to what state of domestic economy suggests
- Secondary finding – No relationship between shocks to profit-shifting activity of MNEs and domestic economy



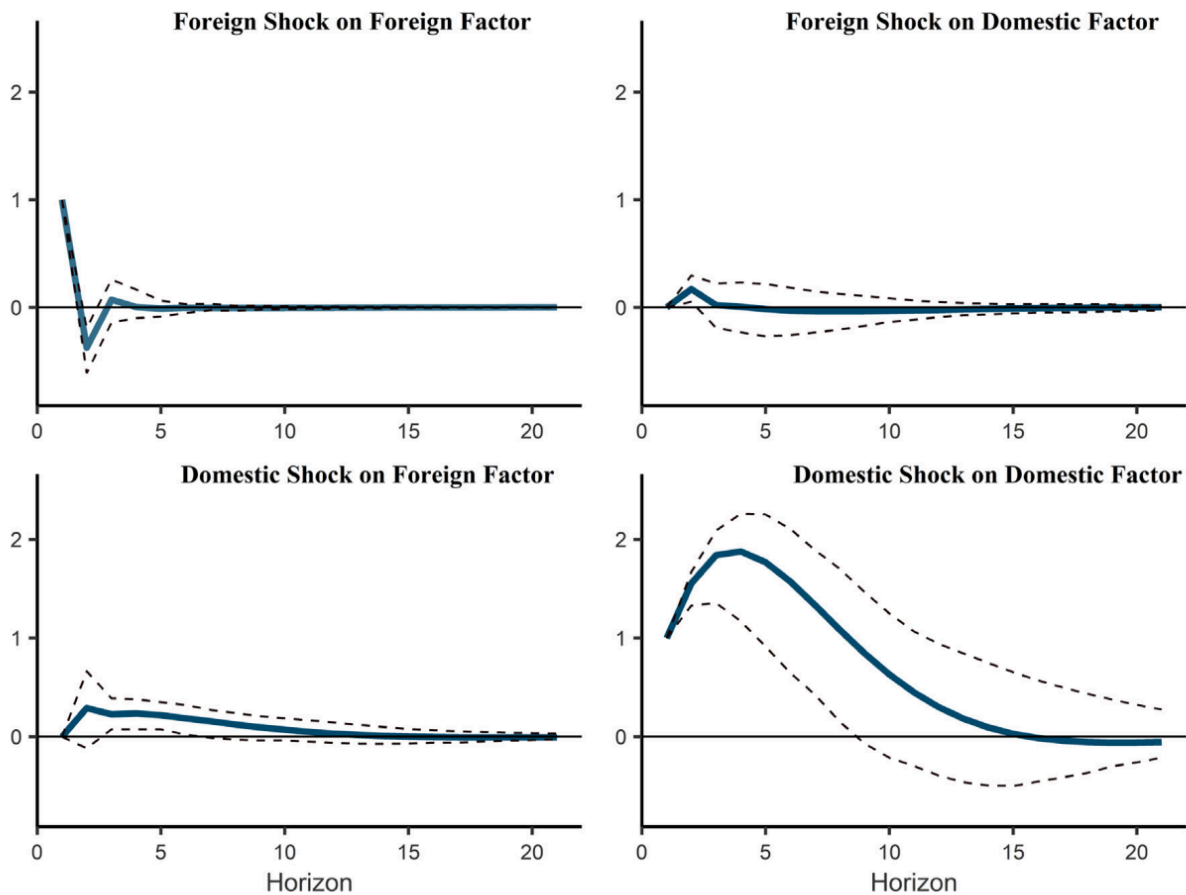
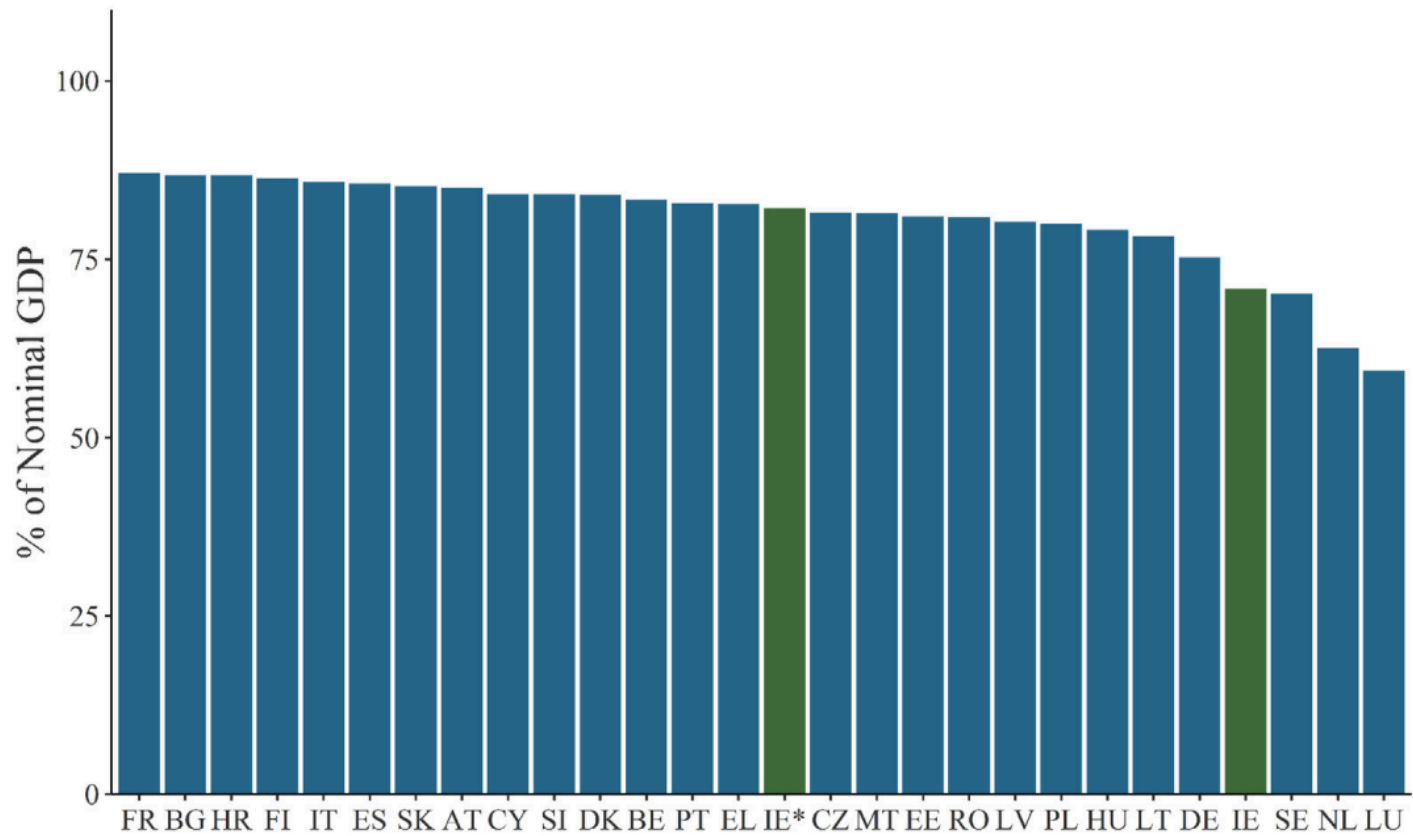


Fig. 3. IRFs for reduced-form shocks to foreign and domestic factors. *Note:* The dashed lines represent bootstrapped 95% confidence intervals.

Panel B: EU Budget Contributions by country in 2018



Research Topics III: Environment/Trade

Global shipping emissions undermine extent to which trade fuels welfare gains

How do new logistical/queuing systems influence emissions?

My work:

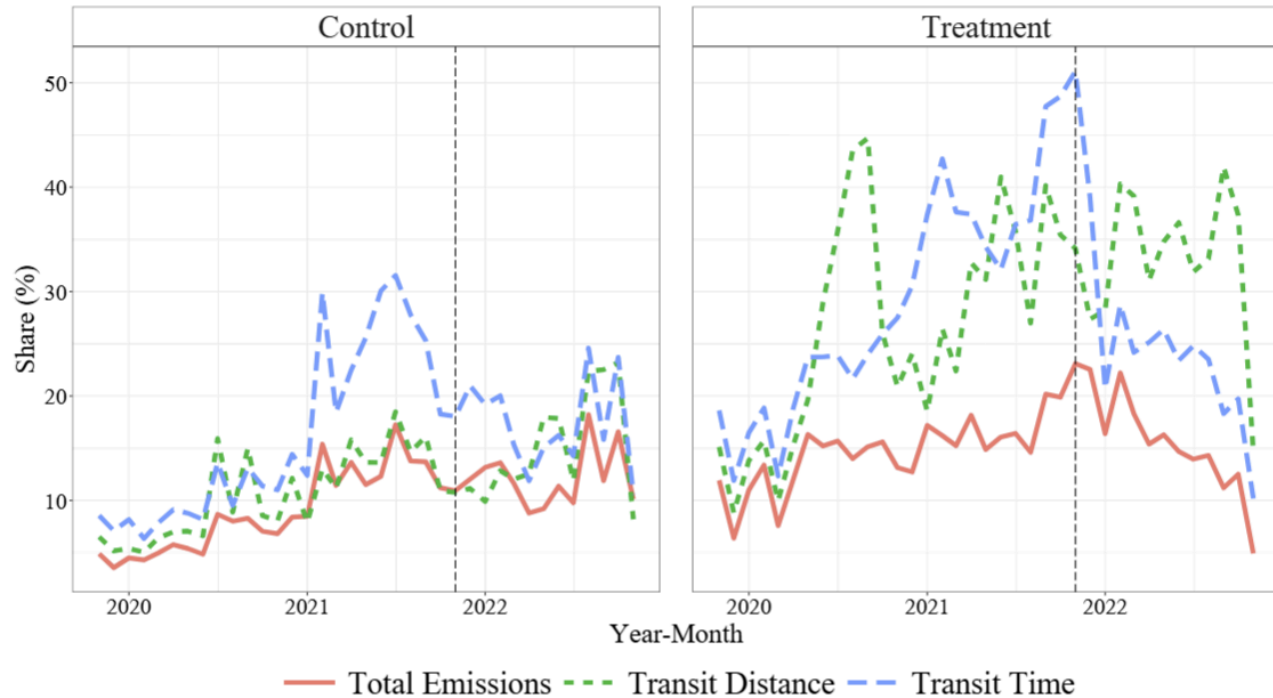
- Considers the new queue system at LA/LB which assigns queue positions based on vessels' estimated time of arrival
- Prior system assigned queue positions upon reaching within 20 nautical miles of LA/LB, incentivized speeding
- How did new system influence global and local emissions?

Table. 6. Difference-in-Difference Estimates – Global Emissions

	CO ₂ Emissions (1)	Em. per Hour (2)	Em. per Knot (3)
Post-Period	-0.2044*** (0.0718)	0.0842 (0.1033)	-0.1270 (0.0819)
Treatment	0.1906* (0.1021)	-0.0238 (0.1843)	-0.1679* (0.0999)
DiD	-0.1107** (0.0456)	-0.0644 (0.0677)	-0.1310** (0.0509)
Vessel-Voyage FE	✓	✓	✓
Year FE	✓	✓	✓
Month FE	✓	✓	✓
Observations	2,142	2,142	2,142
R ²	0.90	0.75	0.71

Note: ***, 0.01, **, 0.05, *, 0.1. Standard-errors are robust to clustering within vessel-voyage lanes of transport service. Each observation is a distinct global transit experience of a vessel arriving on the US west coast between Nov 2019 and Nov 2022. I filter for vessel voyages that maintained the same international trade routes pre- and post-policy and drop any observations that lack matching vessel movement data. To limit outlier distortions, I exclude any voyages with emissions less than the 25th percentile minus three times the interquartile range (75th percentile - 25th percentile) or higher than the 75th percentile plus three times the interquartile range (Davies and Jeppesen, 2015).

New system delivers per-vessel emission reductions of approximately 10 per cent



Driven by a reduction of time spent queuing/idling in local area.

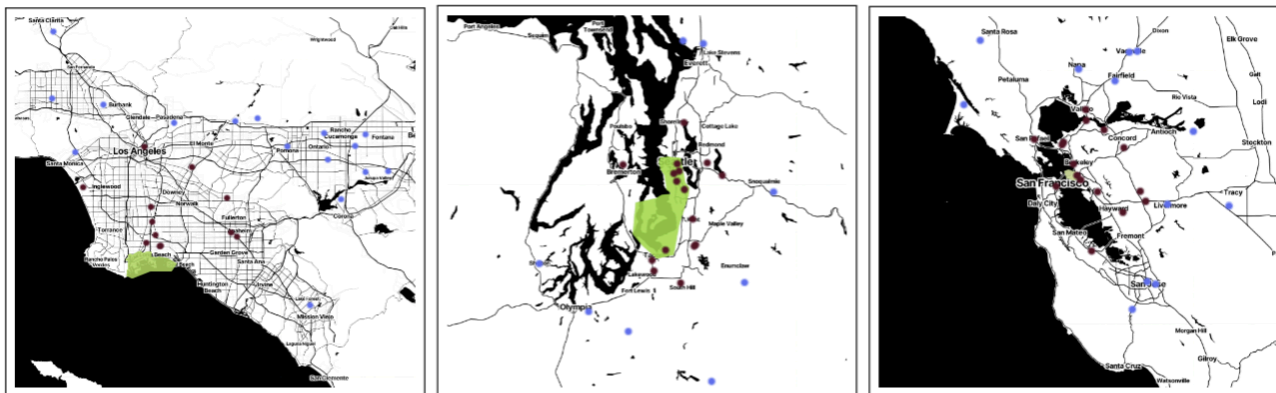


Figure 6. AQI Monitor Local Sites

Local air quality monitors suggest that air quality **deteriorated**

- Within 25-mile radius vs outer 25-50 mile ring
(difference out shocks across LA)
- Within 25-mile LA/LB vs Within 25-mile control ports
(difference out shocks across ports)

Same magnitude of effect across both DiDs.

Table. 10. Difference-in-Difference, Commercial Activity

Dep. Variable:	Trade (USD) (1)	Trade (KG) (2)	Loaded Containers (3)	Total Containers (4)
Post-Period	-0.1077 (0.1596)	-0.1662 (0.1164)	-0.2055*** (0.0459)	-0.2146*** (0.0560)
Treatment	1.019*** (0.0392)	0.7442*** (0.0312)	1.041*** (0.0124)	1.126*** (0.0123)
DiD	0.0864 (0.0849)	0.1310* (0.0674)	0.1575*** (0.0249)	0.2192*** (0.0253)
Month-Year FE	✓	✓	✓	✓
Observations	600	600	600	600
R ²	0.62	0.59	0.95	0.96

Note: ***: 0.01, **: 0.05, *: 0.1. White Standard-errors, robust to the presence of heteroskedasticity. Each observation is a year-month of containerized trade outcomes at the port level between January 2012 and June 2024. ‘Post-Period’ is equal to 1 for dates after November 2021. ‘Treatment’ is equal to 1 for ports of San Pedro Bay. The relevant control group consists of the ports of Oakland, Seattle, and Tacoma. Dependent variables are in log-form. ‘Trade (USD)’ represents the real value of containerized exports and imports, deflated using the St Louis FED’s Consumer Price Index for All Urban Consumers: All Items Less Food and Energy in U.S. City Average. ‘Trade (KG)’ represents the total kilogram weight of containerized imports and exports. ‘Loaded Containers’ uses bilateral loaded container traffic flows reported directly by individual ports as a measure of market share/throughput. ‘Total Containers’ includes bilateral port flows of both loaded and empty containers.

Relative to other west coast ports, LA/LB experienced a surge in trade volume

International Economics:

Key Facts

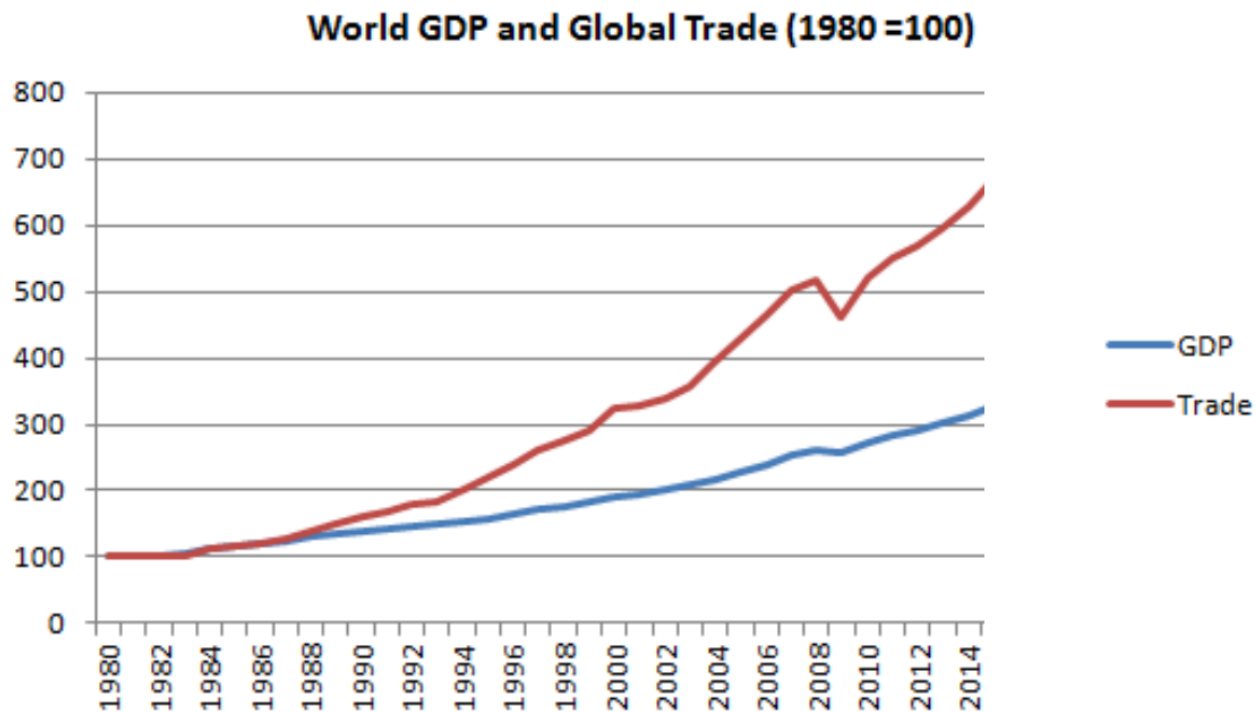
International Trade

Trade Theory $\leftrightarrow$ Trade Empirics

- Trade Theory $\rightarrow$ Testable implications
- Improved data that reveals new facts about trade $\rightarrow$ Theory

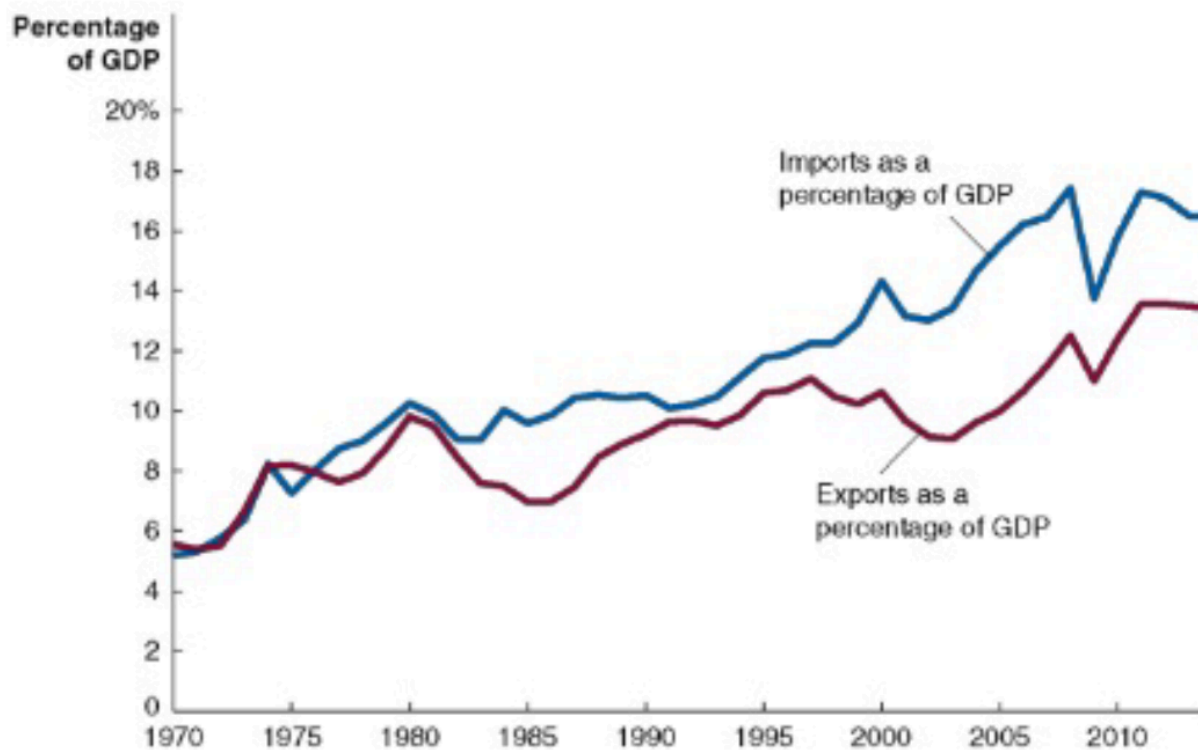
International Trade

Rapid Growth of World Trade ... Faster than World GDP



Source: IMF World Economic Outlook (2012)

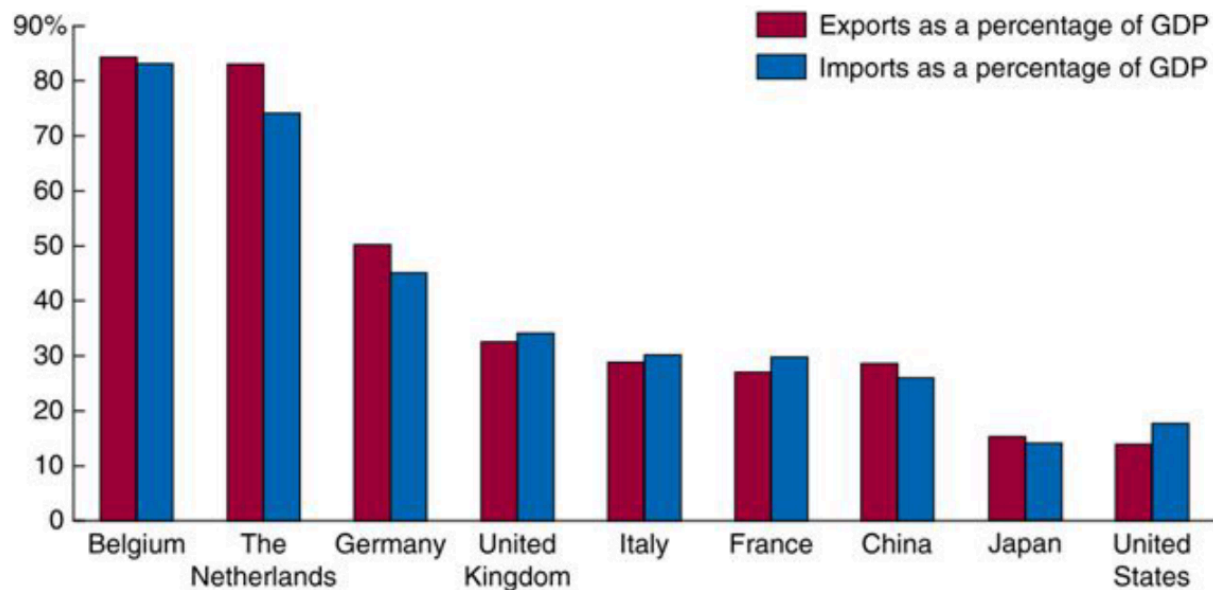
International Trade



Source: Hubbard and O'Brian (6th ed.)

Ratio of Trade to GDP for the U.S., 1970-2014

International Trade



Cross-Country Differences in Trade Openness/Reliance, 2014

Key Facts (I)

- Trade is important: it grows really fast
- Trade is related to economic growth
- Country size is related to global market integration (trade openness)

International Trade

Main Questions:

- Why do countries trade? What are the main determinants of trade?
- What are the gains from trade?
- Does trade cause income growth?

Part I of the course focuses on theories of international trade.

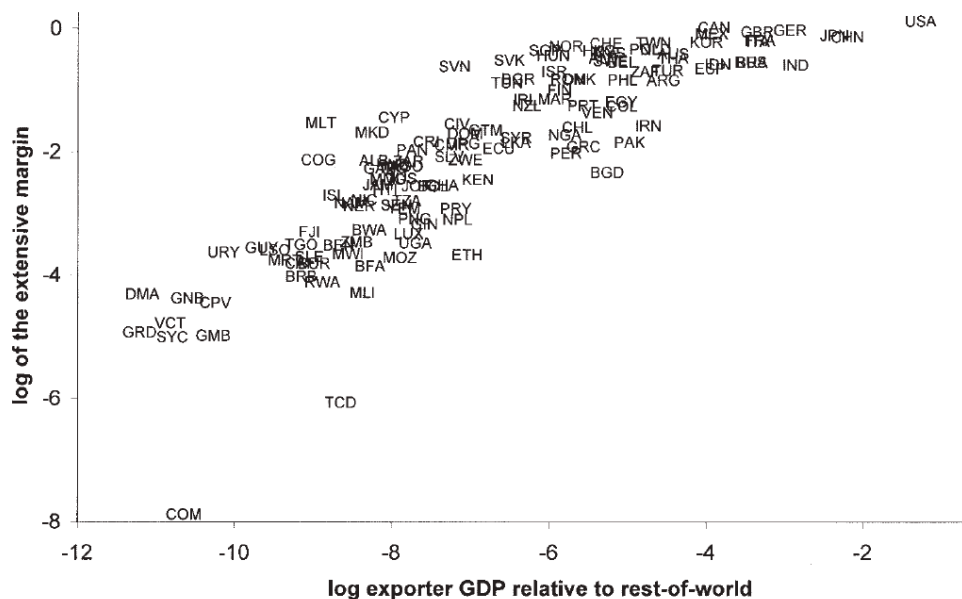
International Trade

Main Questions:

- What goods does each country trade?
- How does trade affect specialization?

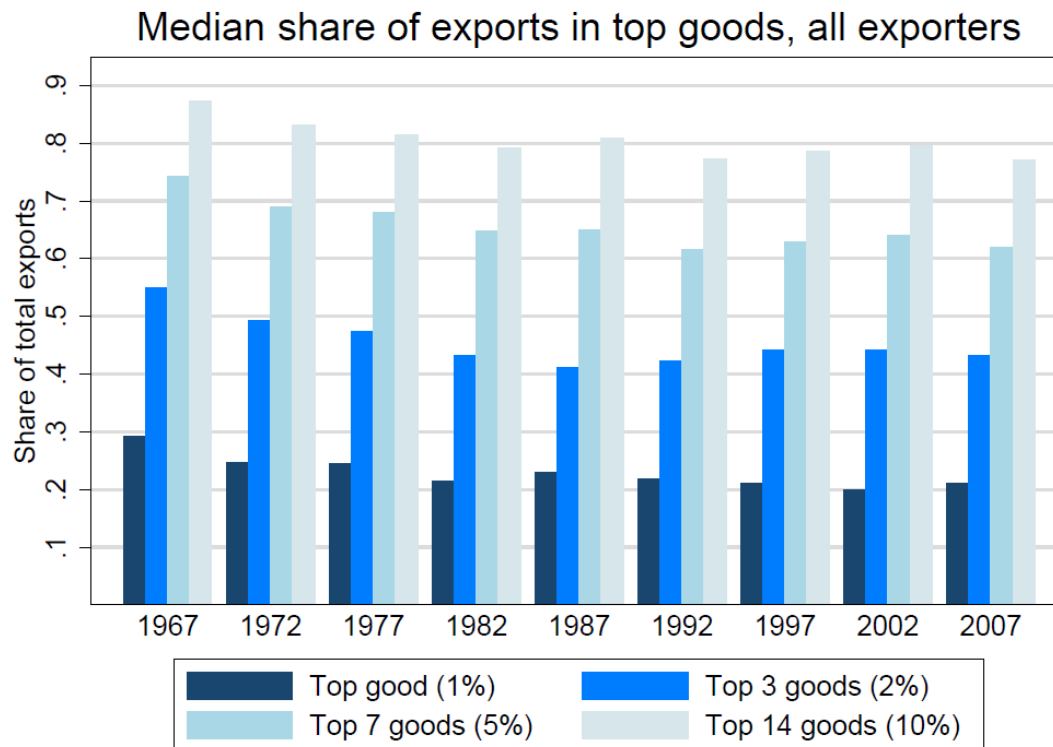
What Goods Do Countries Trade?

Product Range varies with Country Size:



Note: Product extensive margin (y-axis) is a weighted average count of the number of products a country exports in 1995.
Source: Hummels and Klenow (2005)

What Goods Do Countries Trade?



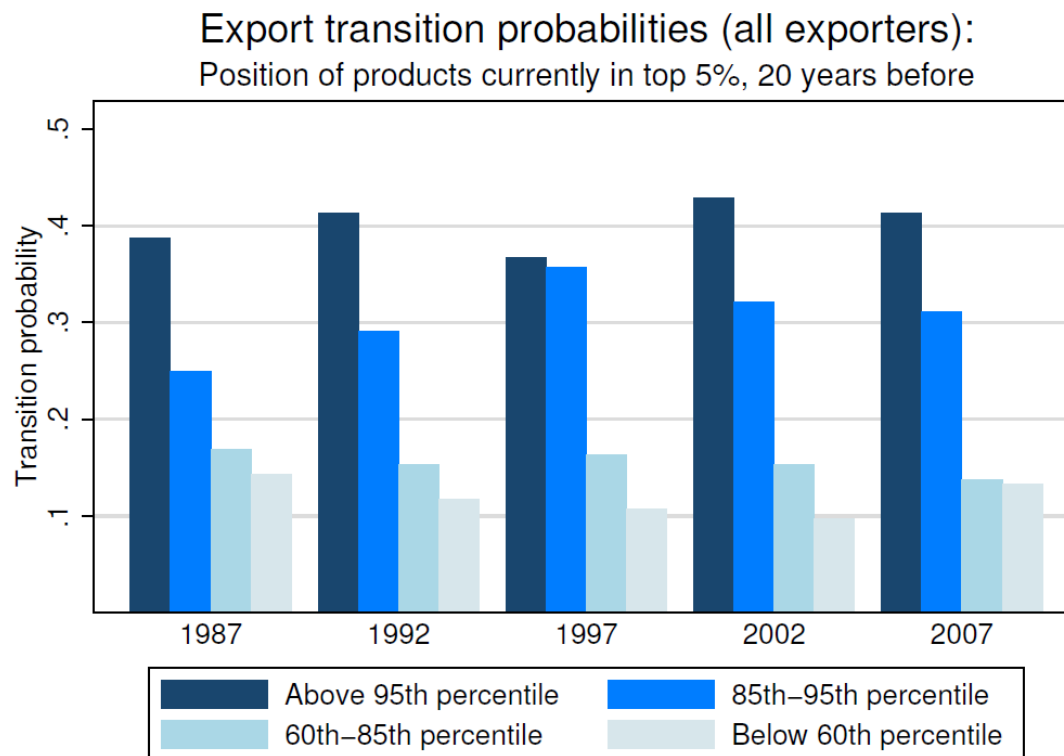
Notes: Sample includes 90 countries and 133 time-consistent industries.
Shares of industry i 's exports in total exports of country c are calculated by country.

Source: Hanson, Lind and Mender (2015)

Signs of Specialization and Comparative Advantage

What Goods Do Countries Trade?

Dynamics in Specialization and Comparative Advantage



Notes: Sample includes 90 countries and 133 time-consistent industries.
Shares of industry i 's exports in total exports of country c are calculated by country.

Source: Hanson, Lind and Mender (2015)

Key Facts (II)

Countries specialize and export core products/sectors

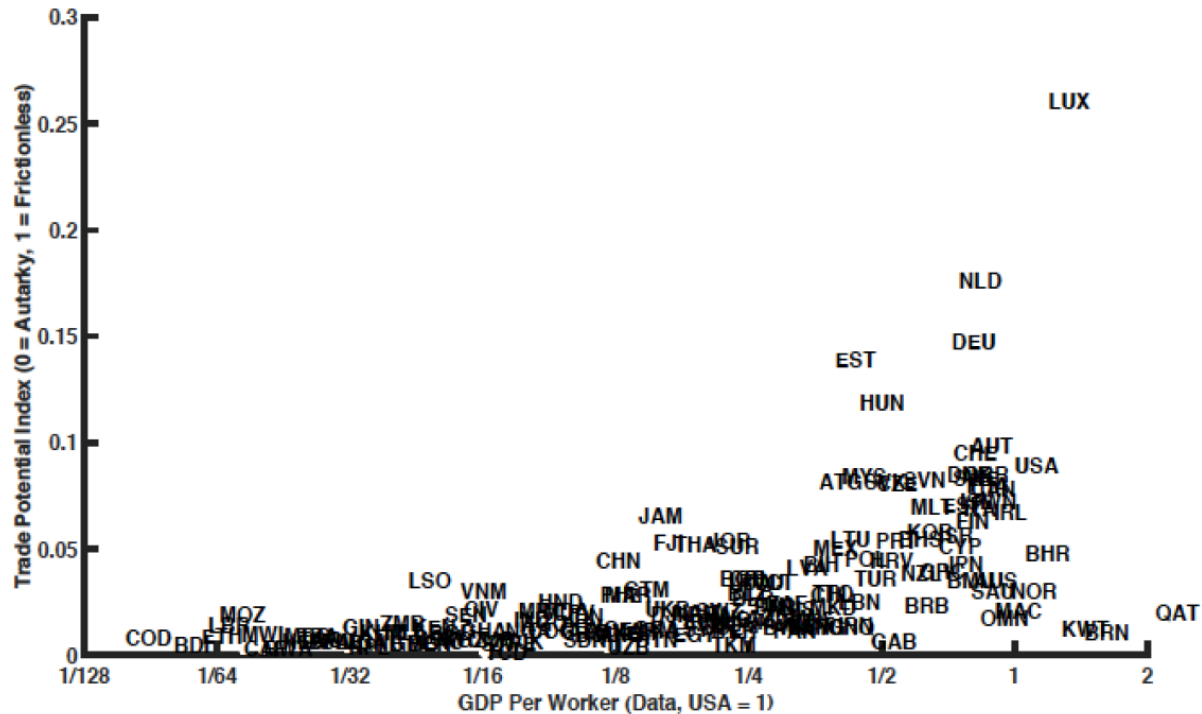
Trading the goods that countries are "best" at → gains from trade

Dynamics in top ranking of export products

Factors other than country fundamentals influence specialization?

- knowledge diffusion?
- foreign direct investments (FDI)?
- cross-border migration?

Barriers to Trade



Trade Potential Index = index capturing a country's position between autarky (index = 0) and frictionless trade (index = 1)

Data for 2005

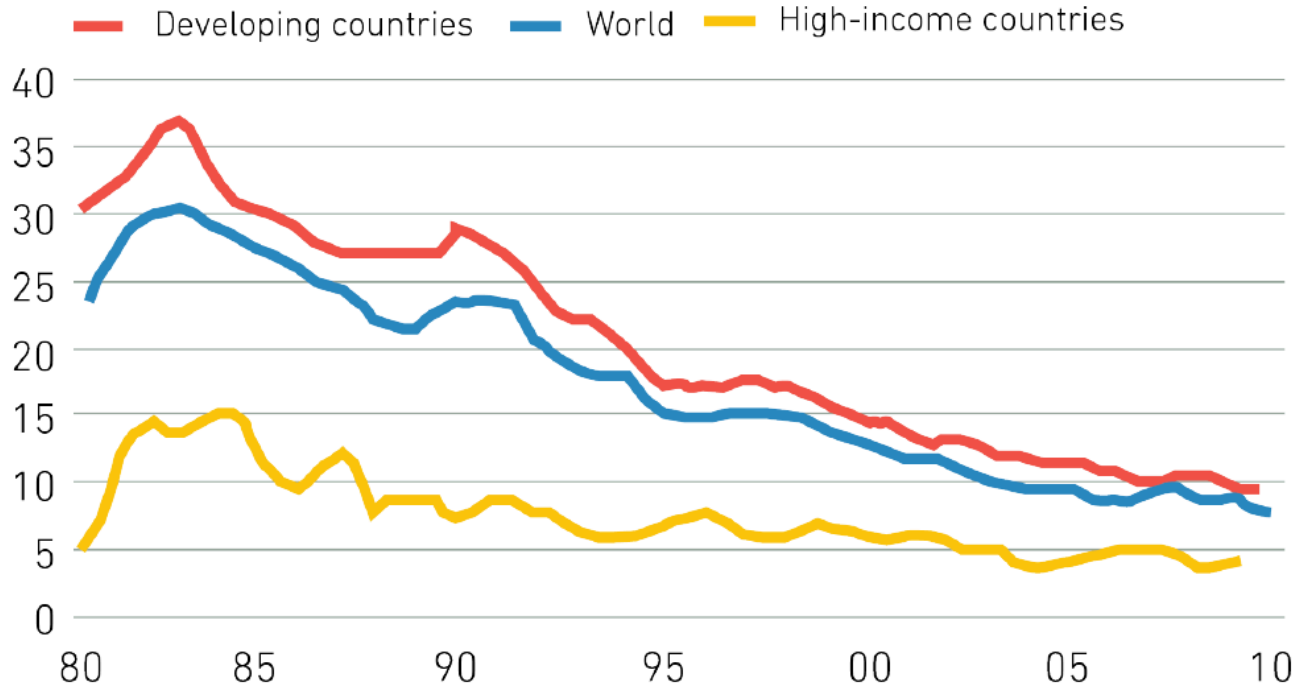
Source: Waugh and Ravikumar (2016)

Trade has been growing rapidly but the world is far from free trade.

Main question:

What barriers hinder international trade?

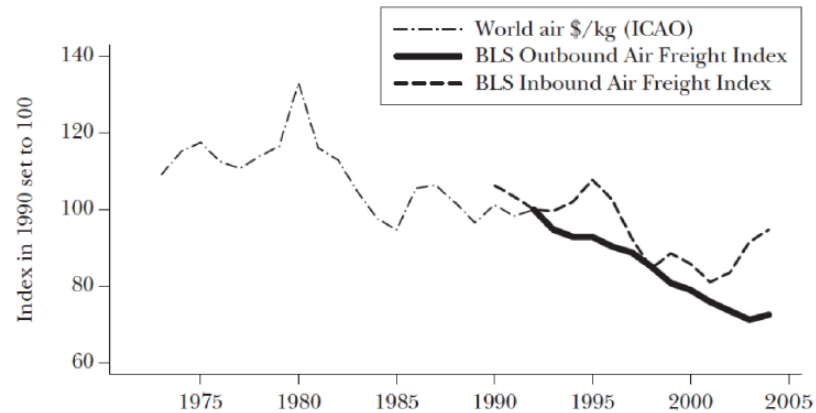
Trends in tariff rates (%)



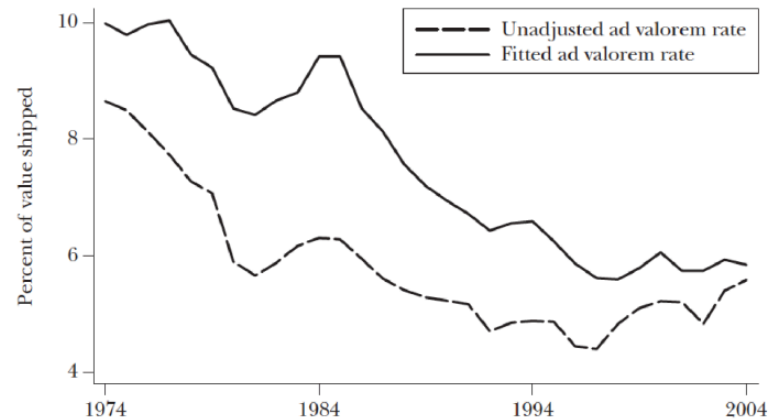
Source: World Bank

Import Tariffs Have Also Dropped Significantly

Air Transport Price Indices



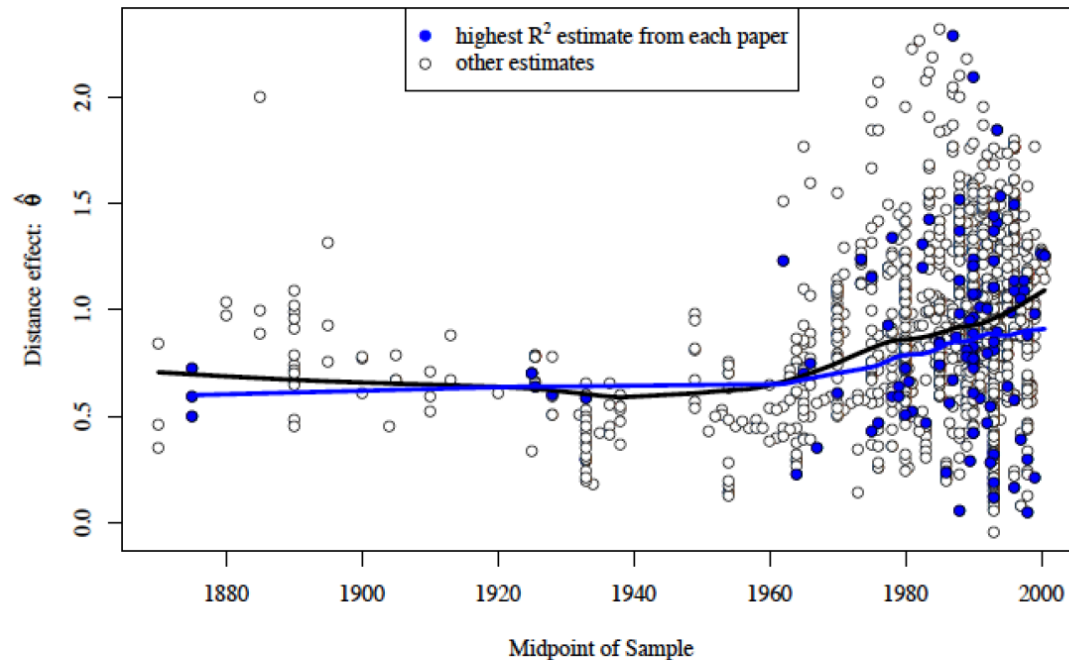
Ad Valorem Ocean Freight



Source: Hummels (2007)

Transport Costs Have Fallen Over Time

$$\text{Regression: } \log \text{Trade}_{ijt} = \alpha + \theta \log \text{Distance}_{ijt} + X'_{ijt} \beta + \epsilon_{ijt}$$



Notes: Meta-analysis of 1450 published papers. Plotted distance elasticity of trade ($\delta \text{Trade} / \delta \text{Dist}$) obtained after controlling for differences in control variables and estimation methods across studies. Source: Disdier and Head (2016)

Distance Puzzle: Persistent Effect of Geographic Distance on Trade Flows

Key Facts (III)

- Explicit trade barriers (e.g., transport costs and tariffs) have decreased over time
- Hidden barriers to trade still play an important role

Identifying and measuring trade costs has important policy implications.

Firm Level Export Activity

So far... facts about country level or industry-level trade

Zooming-in:

- firm level decision to participate in international trade

Main questions:

- Who participates in global markets?
- What are the characteristics of exporting firms?

Firm Level Export Activity

Exporting By U.S. Manufacturing Firms, 2002

<i>NAICS industry</i>	<i>Percent of firms</i>	<i>Percent of firms that export</i>	<i>Mean exports as a percent of total shipments</i>
311 Food Manufacturing	6.8	12	15
312 Beverage and Tobacco Product	0.7	23	7
313 Textile Mills	1.0	25	13
314 Textile Product Mills	1.9	12	12
315 Apparel Manufacturing	3.2	8	14
316 Leather and Allied Product	0.4	24	13
321 Wood Product Manufacturing	5.5	8	19
322 Paper Manufacturing	1.4	24	9
323 Printing and Related Support	11.9	5	14
324 Petroleum and Coal Products	0.4	18	12
325 Chemical Manufacturing	3.1	36	14
326 Plastics and Rubber Products	4.4	28	10
327 Nonmetallic Mineral Product	4.0	9	12
331 Primary Metal Manufacturing	1.5	30	10
332 Fabricated Metal Product	19.9	14	12
333 Machinery Manufacturing	9.0	33	16
334 Computer and Electronic Product	4.5	38	21
335 Electrical Equipment, Appliance	1.7	38	13
336 Transportation Equipment	3.4	28	13
337 Furniture and Related Product	6.4	7	10
339 Miscellaneous Manufacturing	9.1	2	15
Aggregate manufacturing	100	18	14

Characteristics of Exporting Firms

Regression: $\log Y_{it} = \beta_0 + \beta_1 \text{Exporter}(0/1) + \epsilon_{it}$

	Exporter Premia		
	(1)	(2)	(3)
Log Employment	1.19	0.97	.
Log Shipments	1.48	1.08	0.08
Log Value Added per Worker	0.26	0.11	0.10
Log TFP	0.02	0.03	0.05
Log Wage	0.17	0.06	0.06
Log Capital per Worker	0.32	0.12	0.04
Log Skill per Worker	0.19	0.11	0.19
Additional Covariates	None	Industry Fixed Effects	Industry Fixed Effects, Log Employment

Notes: Notes: Data are for 2002 and are from the U.S. Census of Manufactures. All results are from bivariate OLS regressions of firm characteristic in first column on a dummy variable indicating firm's export status. Columns two and three include industry fixed effects and industry fixed effects plus log firm employment, respectively, as additional controls. Total factor productivity (TFP) is computed as in Caves et al (1982). Capital and skill per worker are capital stock and non-production workers per total employment, respectively. All results are significant at the 1 percent level.

Source: Bernard, Jensen, Redding and Schott (2007)

Characteristics of Exporting Firms

Share of Exporting Firms						
Number of Products	Number of Countries					All
	1	2	3	4	5+	
1	40.4	1.2	0.3	0.1	0.2	42.2
2	10.4	4.7	0.8	0.3	0.4	16.4
3	4.7	2.3	1.3	0.4	0.5	9.3
4	2.5	1.3	1.0	0.6	0.7	6.2
5+	6.0	3.0	2.7	2.3	11.9	25.9
All	64.0	12.6	6.1	3.6	13.7	100

Share of Export Value						
Number of Products	Number of Countries					All
	1	2	3	4	5+	
1	0.20	0.06	0.02	0.02	0.07	0.4
2	0.19	0.12	0.04	0.03	0.15	0.5
3	0.19	0.07	0.05	0.03	0.19	0.5
4	0.12	0.08	0.08	0.04	0.27	0.6
5+	2.63	1.23	1.02	0.89	92.2	98.0
All	3.3	1.5	1.2	1.0	92.9	100

Source: Bernard, Jensen, Redding and Schott (2007)

Key Facts (IV)

- Exporting is rare: few firms export + small share of output exported
- Exporting firms are "better" than domestic firms
(e.g., sales, productivity, employment, etc.)
- Most exporters enter few foreign markets and sell few products
- Top exporters responsible for most of trade by value

Firm heterogeneity plays an important role in trade models.

"Exceptional" Firms Are Multinational Corporations

TABLE 1—PRODUCTIVITY ADVANTAGE OF MULTINATIONALS
AND EXPORTERS

Multinational	0.537 (14.432)
Nonmultinational exporter	0.388 (9.535)
Coefficient difference	0.150 (3.694)
Number of firms	3,202

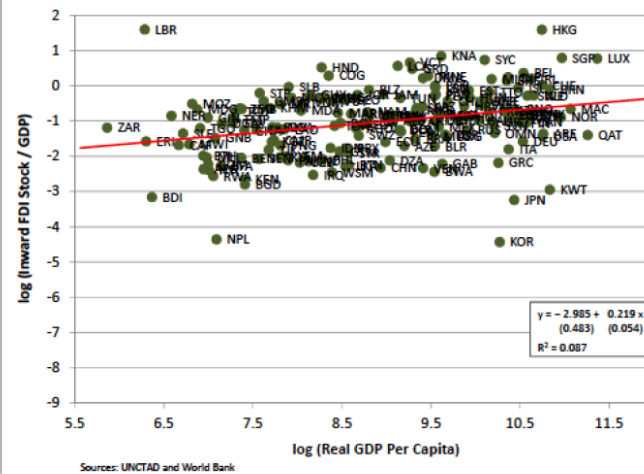
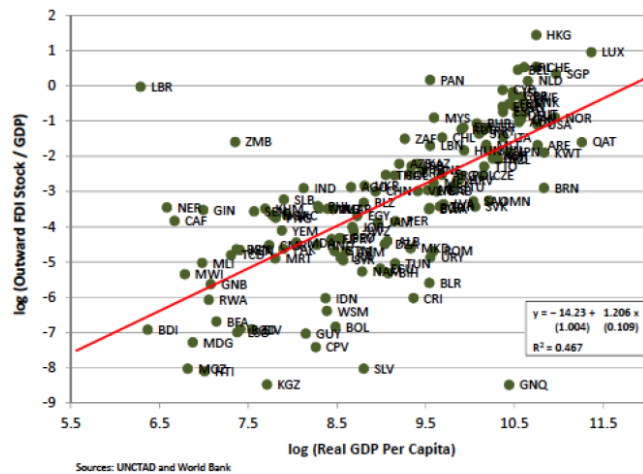
Notes: *T*-statistics are in parentheses (calculated on the basis of White standard errors). Coefficients for capital intensity controls and industry effects are suppressed.

Source: Helpman, Melitz and Yeaple (2004).

Multinational Firms

Main Questions:

- Why do some firms become multinational? What determines FDI?
- Where do multinationals set affiliates?
- What determines location choices?
- How does multinational production affect:
 - host economies?
 - home economies?



Note: Outward FDI stocks are normalized by the GDP size of the sending country (left graph). Similarly, inward FDI stocks are normalized by the GDP size of the receiving country (right graph). Thus, the correlation between FDI and per-capita GDP is not driven differences in country sizes.

Source: Antras and Yeaple (2013)

Multinational activity is concentrated in developed countries

Table 2. Destination of Affiliate Sales by Industry

	Host Country	Other Foreign	United States
Total Manufacturing	55	34	11
Textile and Apparel	45	35	19
Metals and Minerals	60	32	8
Chemicals and Plastics	58	36	6
Machinery	49	36	15
Computers and Electronics	40	43	16
Electronic Equipment	47	40	13
Transport Equipment	47	35	19
Other	66	26	8

Source: 2009 Benchmark Survey of U.S. Direct Investment Abroad, BEA.

Source: Antras and Yeaple (2013)

Foreign affiliates send little output back home

Part I: Trade Theory

1) Basic Trade Models: Why do countries trade? What determines specialization?

Part II: Topics in Trade

2) Distributional Consequences of Trade

What are the consequences of opening up to trade?

For firms and industries? For workers and local labor markets?

3) Multinational Production

Why do firms invest abroad? How do they choose where to locate?

4) Trade Agreements & Optimal Tariffs: Why do countries set protectionist policies?

Part III: Global Finance

We'll open up this can of worms further down the line.

Theories of International Trade

Theories

Two strands within the field of international trade:

1) **Neoclassical models of trade** ('old' theories):

→ Ricardian model

→ Heckscher-Ohlin model

2) **'New' trade theory:**

→ Krugman model

→ Melitz model